

Capital Flow — Product DNA

Snapshot, not living truth. Written 2026-08-04 as a one-time product/business reference for whoever (human or AI) picks up work on this project next. It captures *what the product is and why*, not implementation detail (see [README.md](#) for tech stack, project structure, env vars) and not anything that changes on its own schedule (exact prices, vendor account status, current metrics) — those live in the code and the actual dashboards, not in a markdown file that will silently go stale. If something below contradicts the live code, the code is right.

What this is, in one line

A real-time stock volume scanner for swing traders: it finds stocks trading on unusually high volume — the kind of move that often means real money (institutions, not retail noise) is quietly building a position before the price itself reacts — and hands the trader that information before it's obvious to everyone else.

The problem it solves

Retail traders normally notice a stock *after* it's already moved — a headline, a trending ticker, a friend's tip. By the time they see it, the early move is over. Unusual volume is one of the few signals that shows up *before* the price catches up, but scanning the whole market for it by hand is impractical. Capital Flow automates that scan and surfaces the handful of tickers actually worth a trader's attention, continuously, without them having to watch the market all day.

The product, feature by feature

Capital Flow (the main scanner) — the core feature and the app's namesake. Scans a configurable universe (S&P; 500 / NASDAQ 100 / by sector / everything) for stocks trading at some multiple of their average volume, with filters (min ratio, market cap, price, volume). Every result carries price, volume ratio, RVOL (time-of-day-adjusted volume pace), a 7-day sparkline, float/short-interest, and a link into a chart, live news, and a historical-spike comparison ("has this stock spiked like this before, and what happened after").

MA Scanner — finds stocks trading near a chosen moving average (SMA 9/20/50/150, daily or weekly), for traders whose setup is technical-level-based rather than volume-based.

Hot Sectors (money flow) — a sector-by-sector heatmap across 15 sector ETFs, showing where money is rotating into or out of at a macro level, independent of any single stock.

Watchlist — users star tickers to track them, with live quotes. On top of that sits the alerting system: a per-ticker **volume alert** (fires when RVOL crosses a threshold) or **price alert** (fires when price crosses a target, in either direction) — one-shot, re-armed manually, available from the Watchlist, the main scanner, and the MA Scanner alike so a user never has to leave whatever screen they're on to set one.

Capi — an AI assistant (Elite feature) that explains what a scan result means, in plain language, with memory of the conversation. Deliberately scoped to *explaining* data the app already shows, never to giving investment advice or account-specific answers it isn't actually wired up to know.

Notifications — push notifications and a daily scheduled-scan digest (both Elite features) so a trader doesn't have to have the app open to know something moved.

Admin panel — a single operator's control surface: user list, tier/pilot management, audit log, backup status/trigger, push test, activity stats. Built for one person running the business, not a multi-admin team tool.

The tier model (structure, not current prices)

Three tiers, structured around a simple idea: ****let people feel the full product before asking them to pay for it.****

- **Free** — the complete Elite experience for a 7-day trial from signup, then the account is locked until it upgrades. This is deliberate: the free tier isn't a permanently-crippled demo, it's a full trial with a clock on it.
- **Premium** — unlimited-*feeling* scanning (a generous shared daily pool across every scan type) plus the core analysis features (advanced filters, charts, float/short

data), but no notifications, no Capi, no scheduled scans.

- **Elite** — everything, unlimited, forever: unlimited scans, push notifications, scheduled daily scans, custom watchlist alerts, and Capi.

Both paid tiers are **one-time purchases**, not subscriptions — pay once, keep it. Exact current prices live in `src/constants/tierFeatures.js` and change independently of this document; don't hardcode them anywhere that isn't that one file.

Where the product stands right now (2026-08-04 snapshot)

This snapshot follows a full pre-launch audit pass: security (auth/IDOR/rate-limiting/webhook integrity), scale/reliability (timeouts, background-scan watchdog, crash recovery, real backups with a tested restore path, actual downtime alerting), and correctness (payment/coupon races, price-alert accuracy, DST-safe market-hours math). Every finding from that pass has a corresponding fix and a regression test in the repo. Monitoring (Sentry, PostHog) and uptime (paid Render instance, no more cold-starts) are live and verified working, not just configured. The product is in a genuinely launch-ready state as of this date — "as of this date" being the operative phrase, since that's exactly the kind of fact this file can't promise stays true forever.

The one deliberately-deferred piece from that pass: a short-TTL cache on the live Finnhub price lookup inside the scanner's enrichment phase was identified as a possible further reduction in external API calls, and explicitly **not** implemented — it's the one field in a scan result promised to always be maximally fresh, and the tradeoff wasn't judged worth it. Revisit only if Finnhub rate-limiting becomes a real, observed problem, not preemptively.

How to keep this file useful

Don't add anything here that a script could compute from the live code or a dashboard instead — that's exactly the class of content that quietly rots and then misleads the next reader. When the *product itself* meaningfully changes (a new core feature, a tier

restructuring, a pivot in who this is for), update the relevant section above. When only a number changes (a price, a vendor's plan, a metric), don't — point at where that number actually lives instead.